

WORKSHOP MANUAL

Type GBX-450E Two-Stage Reduction Gearbox

Document	WM-GBX-450E, revision C
Manufacturer	ADLOID GEARWORKS — Industrial Drives Division
Applies to	GBX-450E, all serial numbers
Scope	Procedures 10–40 (disassembly for overhaul). Sections 50–70 not digitised.
Source	OCR of 1998 paper manual, partial 2004 update
Corrections	Service Bulletin SB-2019-04 is authoritative where it conflicts

REVISION HISTORY

Rev	Date	Description	By
A	1998-06-01	First issue (paper)	Eng.
B	2004-11-02	OCR digitisation, procedures 10-40	K.M.
C	2019-04-05	Torque + qty updates, ref SB-2019-04	Doc.

WARNING	Isolate the drive and lock out the input shaft before any work. The unit may weigh up to 42 kg dry. Use a rated lifting eye at the housing boss.
CAUTION	This is an OCR'd document; figure references were not scanned and some torque/quantity values were corrected in SB-2019-04. Cross-check before use.

SECTION 0 — ORIENTATION & CONVENTIONS

Drive end / A-side = the input-pinion side. **Output end / B-side** = the coupling-flange side. **Front** = the cover flange face carrying the locating dowel pins; **rear** = the opposite bolt ring. Bolt rings are identified by their seating face, not by size — the front flange, mid ring and rear ring all use the same M6 hex bolt. When in doubt, refer to the exploded view below.

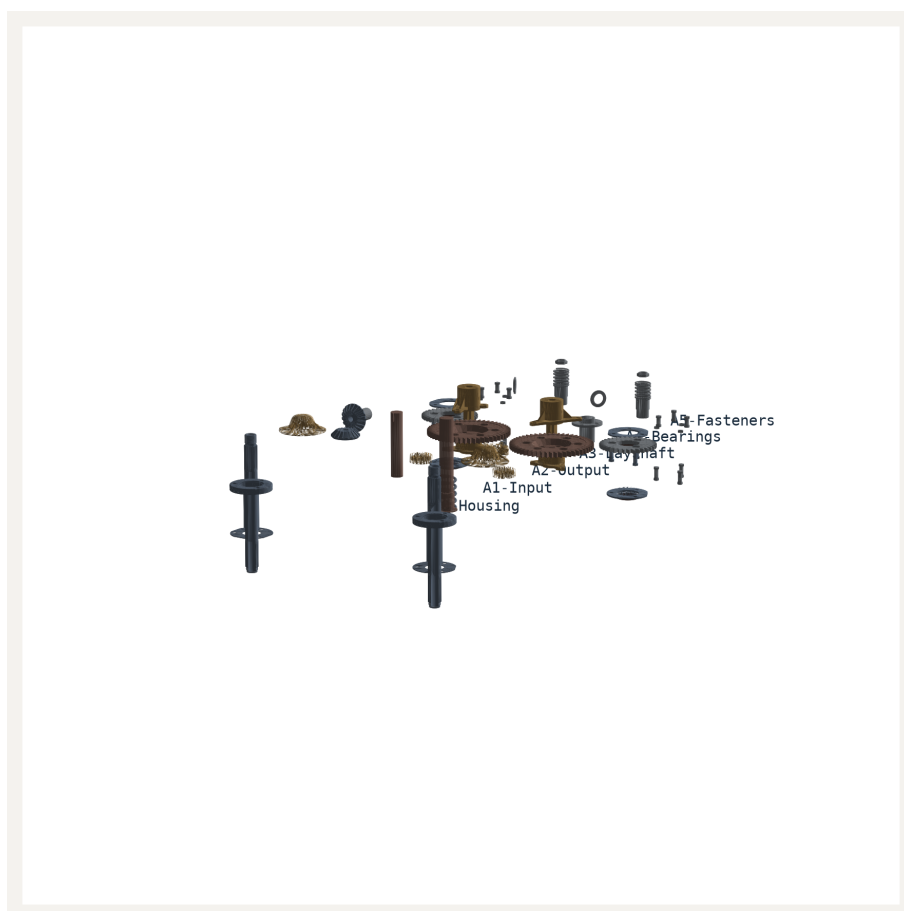


Fig 1-1 — Exploded view (part numbers in IPC-GBX-450E)

PROCEDURE 10 — COVER REMOVAL & INSPECTION

Step 10-10 — Drain & remove breather plug

Place unit on bench, output flange facing operator. Unscrew the breather/filler plug from the top of the housing (17 mm). Allow residual oil to drip for approx. 10 min. Check the sight glass for metal swarf. NOTE: sight glass is not fitted on -E variants.

Step 10-20 — Remove front cover flange bolts

Remove the six M6 hex head bolts on the front cover flange (outer bolt circle) in a criss-cross pattern. Torque on refit: 9.5 Nm (84 lb-in). Do NOT confuse these with the mid-ring bolts -- flange bolts sit on the raised boss.

CAUTION

Slacken flange bolts in a criss-cross pattern to avoid warping the cover. Refit torque 9.5 Nm (84 lb-in).

Step 10-30 — Extract locating dowel pins

Using a pin puller, withdraw all six locating dowel pins from the front cover mating face. Tag each pin with its bore position; pins are matched to their bores from factory honing.

PROCEDURE 20 — DRIVE-END SEAL & INPUT SHAFT SERVICE

Step 20-10 — Remove drive-end front bearing cap

On the drive end (A-side, nearest the input pinion), remove the FRONT bearing retainer cap. Ref. GBX-BC-108. See Fig. 4-2.

Step 20-20 — Remove drive-end rear bearing cap

Rotate the unit and remove the REAR bearing retainer cap on the same (drive) end. Keep shim pack together -- shim thickness sets pinion mesh preload.

NOTE

Keep the shim pack together and with its cap — shim thickness sets pinion mesh preload and is selective on assembly.

Step 20-30 — Free the A-side mid-ring bolts

Slacken and remove both mid-ring retaining bolts on the drive end (mid bolt circle, z-mid row). CAUTION: bolts are grade 8.8 -- do not substitute.

Step 20-40 — Replace drive-end front oil seal

Pry out the radial oil seal (Simmering type) from the drive-end front bore. Fit new seal GBX-OS-124-B flush to 0.5 mm below face; lip grease: Klueber GL-261. Old O-ring must be discarded.

Step 20-50 — Withdraw input shaft & pinions

Support the input shaft and withdraw it together with both input pinions toward the drive end. Do not let the pinion teeth strike the case web.

Step 20-60 — Remove input circlips & thrust washers

Remove the internal circlips (DIN 472) and both bronze thrust washers from the upper bores. Circlip pliers A-2 required.

PROCEDURE 30 — LAYSHAFT & GEAR CLUSTER OVERHAUL

Step 30-45 — Remove oil pump drive [DELETED]

Not applicable to the GBX-450E series — no oil pump fitted. Step number retained for continuity (SB-2019-04).

Step 30-10 — Remove rear ring bolts

Remove the six hex bolts of the rear bolt ring (rear cover row, opposite the front flange). Refit torque 9,5 Nm. One bolt is fitted with a bonded washer -- note its position.

Step 30-20 — Remove socket cap screws

Remove all six DIN 912 socket head cap screws (5 mm hex) from the case split line. Refit torque: 13 Nm / 115 lb-in / 132 kpcm.

Step 30-30 — Split housing halves

Using jacking screws in the cast bosses, separate the two housing halves evenly. NEVER insert a screwdriver in the split line. Lift the upper half clear of both layshafts.

WARNING

Never insert a screwdriver in the split line to separate the housing halves — use the jacking screws in the cast bosses only.

Step 30-40 — Lift out layshaft assemblies

Lift both layshaft (countershaft) assemblies out of the lower half. Keep A-side and B-side shafts separate; they are NOT interchangeable after run-in.

Step 30-50 — Strip cluster & helical gears

Press the cluster gear blocks and all helical reduction gears off their shafts. Inspect tooth flanks for pitting (max. allowable: DIN 3979 class 2). Mark gear-to-shaft clocking.

Step 30-60 — Replace needle bearings

Drive out all six needle roller bearing cages from the case webs using drift T-4408 and press in new cages to a depth of 2.5 mm. Lube with assembly paste before fitting.

Step 30-70 — Renew snap rings, shims & sleeves

Renew both external snap rings and the adjusting shim, and check both distance sleeves for scoring (replace if > 0.05 mm). Shim pack sets end-float 0.10-0.25 mm.

PROCEDURE 40 — OUTPUT END SERVICE**Step 40-10 — Remove coupling flange & key**

Lock the output shaft and remove the coupling flange from its taper. Extract the parallel shaft keys (DIN 6885) from both keyways -- use key puller, do not lever against the seal face.

Step 40-20 — Withdraw output shaft & spur gear

Withdraw the output shaft together with both output spur gears toward the B-side. Support the gear rim; combined mass approx. 6,2 kg.

Step 40-30 — Replace B-side bearings & seals

On the output end, extract both cylindrical roller bearings and both radial oil seals, then refit the output-end bearing caps (front and rear) with new gaskets. Heat bearings to 80 degC max. for fitting; never use open flame.

CAUTION

Heat bearings to 80 °C max for fitting. Never use an open flame; induction or oil bath only.

END OF EXTRACT. Sections 50–70 (testing, paint, storage) not digitised. — scanned by K.M., 2004-11-02